ASSIGNMENT#01

SE4095 Social Network Theory and Practice

Core rule. You must choose your own centrality and community-detection methods, but every chosen method must come from the allowed course lists in this assignment. Ground-truth political labels may be used only in the evaluation stage.

Learning outcomes

- Represent a real social system as a directed graph and explain its nodes, edges, direction, and attributes.
- Calculate and interpret suitable centrality measures while handling graph direction, disconnected components, and parameters.
- Use centrality rankings to design a network experiment and study their effect on detected communities.
- Select a community-detection approach that matches the network and justify its assumptions.
- Evaluate communities with a confusion matrix, accuracy, precision, recall, F1 score, and an internal network score.

Dataset

Political Blogs Network. The data represents political blogs as nodes and hyperlinks as directed edges. The GML node attribute named value stores the binary political-orientation label that serves as ground truth only after community detection.

Dataset page: Mark Newman Network Data

Direct download: polblogs.zip

File to use. polblogs.gml can be loaded by NetworkX, igraph, Gephi, or an equivalent graph library.

Data note. Different libraries may report slightly different counts after removing self-loops, duplicate edges, isolates, or disconnected nodes. Do not copy expected counts. Report the counts produced by your documented preprocessing.

Problem statement

Investigate how influence and brokerage appear in the political-blog hyperlink network, whether the link structure recovers the two known political orientations, and how the network's most central blogs affect the stability and quality of those communities.

Assignment workflow

Stage	Work	Main output
1	Dataset and graph construction	Clean graph and data profile
2	Graph structure	Properties and justified preprocessing
3	Centrality selection	Scores, rankings, and interpretation
4	Centrality-informed experiment	Modified graph and comparison baseline
5	Community detection	Detected memberships without label leakage
6	Community evaluation	Metrics, visual evidence, and conclusion

Part A Dataset and graph construction

Marks 15

- Load polblogs.gml and identify the node identifier, political-orientation attribute, and edge meaning. Preserve the orientation label for later evaluation, but do not use it to construct, tune, seed, or select communities.
- Determine whether the graph is directed or undirected, weighted or unweighted, simple or a multigraph. Explain why that representation matches the data.
- Inspect missing attributes, self-loops, duplicate edges, isolates, and disconnected components. State every preprocessing decision and show graph size before and after cleaning.
- Report the number of nodes, number of edges, density, weakly connected components, strongly connected components, and in-degree and out-degree distributions.
- For every distance-based calculation, state which component and edge-direction convention you use. Explain why the choice is valid.

Required evidence

- A compact data dictionary and preprocessing log.
- One degree-distribution figure and one readable network visualization. Sampling or filtering is allowed only for visualization; calculations must use the stated analysis graph.
- A short interpretation linking the observed graph properties to connectivity, homophily, or information flow.

Part B Centrality selection and analysis

Marks 25

Select any two centrality measures from the taught list. The assignment does not prescribe which two to use. Your selection must follow from the graph's direction, structure, and the question you want each measure to answer.

Allowed centrality	Question it can support
Degree, in-degree, or out-degree centrality	Direct connectivity or attention given or received
Eigenvector centrality	Importance through important neighbors
Katz centrality	Influence through paths with an explicit bias term
PageRank	Authority in a directed link network with divided outgoing influence
Betweenness centrality	Brokerage on shortest paths
Closeness centrality	Ability to reach other nodes through short paths

Scope restriction. Do not use a centrality measure outside this list, even as an additional result.

- Write one network question for each chosen centrality and justify why that measure answers it better than the alternatives in the taught list.
- State the formula or computational idea, normalization, parameters, direction convention, and treatment of unreachable nodes. For Katz or PageRank, justify the parameter values. For iterative methods, report convergence handling.
- Compute a score for every node and produce a ranked top-10 table for each measure. Include node name or label, raw score, normalized score if used, and rank.
- Compare the two rankings using a suitable numerical comparison and explain agreements and disagreements. A rank correlation or top-k overlap is acceptable.
- Interpret at least three important nodes in network terms. A high score alone does not prove real-world political importance.

Part C Centrality informed community experiment

Marks 20

Use one selected centrality to design a controlled experiment before evaluating communities. This stage must visibly connect the centrality analysis to the community-detection work.

Choose one selected centrality as the intervention measure and justify the choice.

- Remove a small, justified proportion of the highest-ranked nodes to create a centrality-targeted graph. You decide the threshold; report it clearly and keep enough nodes for meaningful analysis.
- Create a random-removal graph by removing the same number of nodes. Fix and report the random seed.
- Run the same chosen community-detection procedure on the original, centrality-targeted, and random-removal graphs using comparable settings.
- Compare community count and sizes, component structure, and evaluation results across the three graphs. Evaluate each removal graph only on its remaining nodes.

Part D Community detection

Marks 20

Select one community-detection method from the taught list. You must decide which method fits the graph and explain your reasoning. Do not choose by checking which method gives the best ground-truth score.

Allowed method family	Relevant structure from the lectures
Clique-based methods	Clique, relaxed clique, k-core, or Clique Percolation Method
Reachability or similarity methods	k-clique, k-club, k-clan, or clustering based on Jaccard or cosine similarity
Label propagation	Fast neighborhood-label process; results may vary by run
Modularity maximization	Non-overlapping dense groups; Louvain is included
Spectral clustering	Graph Laplacian and eigenvector representation
Hierarchical clustering	Nested grouping represented by a dendrogram
Stochastic block model	Probabilistic blocks with group-dependent connection rates
InfoMap	Communities based on compression of random walks

- Explain whether the method assumes disjoint, overlapping, balanced, dense, hierarchical, or probabilistic communities and whether it supports a directed graph. Document any justified transformation to an undirected graph.
- Report all parameters and the detected number and size distribution of communities.
- If the method is stochastic, run it with at least five recorded seeds and report mean and standard deviation for the evaluation metrics. Use one declared run for visualizations.
- Produce a community-colored network visualization and a table that profiles each major community using the two centralities from Part B.

Part E Community evaluation

Marks 15

After detection is complete, use the node political-orientation label as ground truth. Detected IDs are arbitrary and the number of communities may differ from two, so evaluate unordered pairs of nodes. For overlapping output, treat a pair as belonging to the same detected community when the two nodes share at least one community, and document how unassigned nodes are handled.

Term	Pairwise meaning
True positive	Same detected community and same ground-truth orientation
True negative	Different detected communities and different ground-truth orientations
False positive	Same detected community but different ground-truth orientations
False negative	Different detected communities but same ground-truth orientation

Required external metrics

Metric	Calculation
Accuracy	(TP + TN) / (TP + TN + FP + FN)
Precision	TP / (TP + FP)
Recall	TP / (TP + FN)
F1 score	2 x Precision x Recall / (Precision + Recall)

- Show the pairwise confusion matrix and all four external metrics for the original, targeted-removal, and random-removal graphs.
- Report modularity as the internal structural quality score whenever it is defined for your graph and partition. If communities overlap, explain the modularity variant or state why standard modularity does not apply.
- Do not compare numeric community IDs directly with 0/1 labels. Explain how pairwise evaluation avoids label-permutation problems.
- Discuss what precision and recall mean in this network. Explain one reason why strong structural communities may still disagree with political orientation.

Zero denominator rule. If a denominator is zero, report the metric as undefined and explain the condition. Do not silently replace it with zero.

Part F Reproducibility and reflection

Marks 5

- Keep the notebook runnable from top to bottom and set all random seeds.
- Separate data loading, graph preparation, centrality, experiment, community detection, evaluation, and visualization into clear functions or sections.
- End with a concise conclusion that answers the problem statement and acknowledges limitations, including historical context and the risk of equating hyperlinks with endorsement or influence.

Submission requirements

Item	Format	Required content
Code	Jupyter notebook or Python project	Complete implementation, comments, fixed seeds, and outputs
Report	PDF, 5 to 7 pages excluding appendix IEEE Format only	Decisions, tables, figures, metrics, interpretation, and conclusion
Processed data	CSV, GML, GraphML, or script	Only if preprocessing changes the supplied graph
Readme	Plain text or Markdown	Setup steps, dependencies, and exact execution instructions

- Submit one ZIP file named StudentID_SNTP_Assignment.zip.
- Use only relative paths. The evaluator must not need to edit local file paths.
- Credit every external code fragment or resource and follow the course academic-integrity policy.

Marking rubric

Area	Marks	Assessment focus
Dataset and graph construction	15	Correct graph, transparent preprocessing, graph properties, and visual evidence
Centrality selection and analysis	25	Justified choices, correct implementation, parameters, rankings, and interpretation
Centrality-informed experiment	20	Targeted-versus-random design, reproducibility, and meaningful comparison
Community detection	20	Independent choice from taught methods, correct application, parameters, and community profiles
Community evaluation	15	Pairwise confusion matrix, accuracy, precision, recall, F1, modularity, and interpretation
Reproducibility and reflection	5	Runnable work, clear structure, limitations, and concise conclusion
Total	**100**	

Minimum completion checklist

- The graph model and preprocessing are documented.
- Exactly two taught centralities are selected and justified.
- Top-10 centrality rankings and their comparison are included.
- A targeted-removal and equal-size random-removal experiment is completed.
- One taught community-detection method is selected without using labels.
- The pairwise confusion matrix, accuracy, precision, recall, and F1 score are correct.
- Modularity is reported or its non-applicability is justified.
- Results for the original and both removal graphs are compared.
- The submission runs from the supplied instructions and uses fixed seeds.

Dataset citation

Adamic, L. A., and Glance, N. (2005). The Political Blogosphere and the 2004 U.S. Election: Divided They Blog. Proceedings of the 3rd International Workshop on Link Discovery.

Source and access: https://public.websites.umich.edu/~mejn/netdata/